MANUFACTURING ANALYTICS FOR ENGINEERS · SECTION 9 CAPSTONE

The plant manager's five questions

Engxtech Manufacturing, April to September 2026 · about 50 minutes · free Power BI Desktop

The situation

Engxtech's plant manager has a review with the operations director next week. She has asked you for one page and one short email that answer five questions, each with a number. Use the model and measures you built in sections 5 to 8.

The five questions

1. Which line and which shift lose the most OEE, and on which factor (availability, performance or quality)?
2. Which failure mode should maintenance attack first, measured by repair hours?
3. Which machine should quality fix first, and what is the evidence?
4. Is energy per good unit getting better or worse over the six months?
5. Which days were abnormal, and what happened on them?

What to hand in

File	What it contains
Section09_Capstone.pbix	Your Section 8 file plus one new page, built from blank, that answers all five questions. Practice pages hidden.
Section09_Capstone.pdf	That page exported with File › Export › Export to PDF, slicers cleared.
Email (five sentences)	One finding per question, each with its number, and one recommendation.

Check your own work

Check	Pass when
Answers	All five questions answered on one page, each with its number
Correct numbers	Your numbers match the answer slide in the Section 9 deck
Readable in five seconds	KPIs on top; every visual title states its finding
Honest charts	Axes start at zero or say they don't; limits shown on the trend
Shareable	Practice pages hidden; the PDF is one clean page
The email	Five sentences, one recommendation, no jargon

> **Hints.** Reuse your measures: OEE, Availability, Speed %, Quality, Breakdown hours, Cumulative %, Cpk, Wh per good unit, UCL and LCL. Put the five answers in the order of the questions, top-left to bottom-right. Don't open the answer slides until your email is written.

Practice data: 02_Learner_Data. Finished files for checking: 06_PowerBI_Files. Engxtech Manufacturing is a fictional plant built for this course.